



Mansoura University
Faculty of Computers and Information
Fall 2023-2024



DECISION SUPPORT SYSTEMS

Grade: 4TH YEAR (GENERAL)

Prepared by:

Ibrahim El-Hasnony

QUIZ I

- 1)is a process by which organizational goals are achieved using resources
(Decision, Modelling, System ,Management)
- 2)is a choice made between alternative courses of action in a situation of uncertainty.
(Decision, Decision making, Decision support systems, Information system)
- 3)measures the extent to which the system meets its objectives.
(reliability, effectiveness, efficiency, completeness)
- 4) TPS may work inmode, processing accumulated transactions at a single time later
(batch, online, real time, immediate)
- 5) DSS support all organization levels except.....level.
(strategic, operational, management, tactical)

QUIZ I

- 1) Management reporting systems report on the past and the present, rather than projecting the future ()
- 2) Executive information systems support top managers with conveniently displayed summarized information, customized for them ()
- 3) The core management activates for planning in the management level of organization ()
- 4) Decision is a cognitive process that results in the selection of a course of action among several alternative scenarios ()
- 5) Pareto chart helps you to identify all possible causes of the problem ()
- 6) Forward Analysis Sensitivity Models known as goal seeking analysis ()

COURSE DETAILS

Course Syllabus

Chapter 1: Introduction to DSS

Chapter 2: Decision Making Concepts

Chapter 3: Decision Making Technologies and Environment

Chapter 4: Model Management

Chapter 5: Data Management System and Dialogue Management

Chapter 6: Designing and Developing Decision Support Systems

Chapter 7: Executive Support Systems

Chapter 8: Automated Decision Systems and Expert Systems

Chapter 9: Knowledge Management and Collaborative Systems

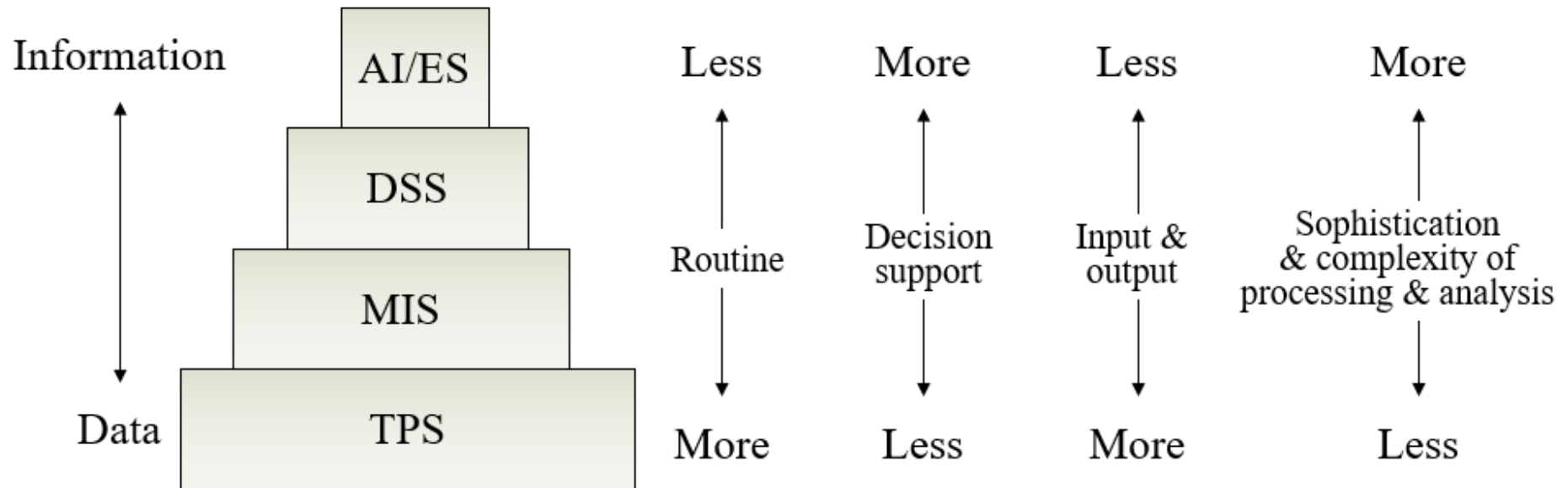
Chapter 10: DSS Intelligent Systems & Tools and Techniques

Chapter 11: Business Intelligence: Data Warehousing, Data Acquisition, Data Mining, Business Analytics, and Visualization

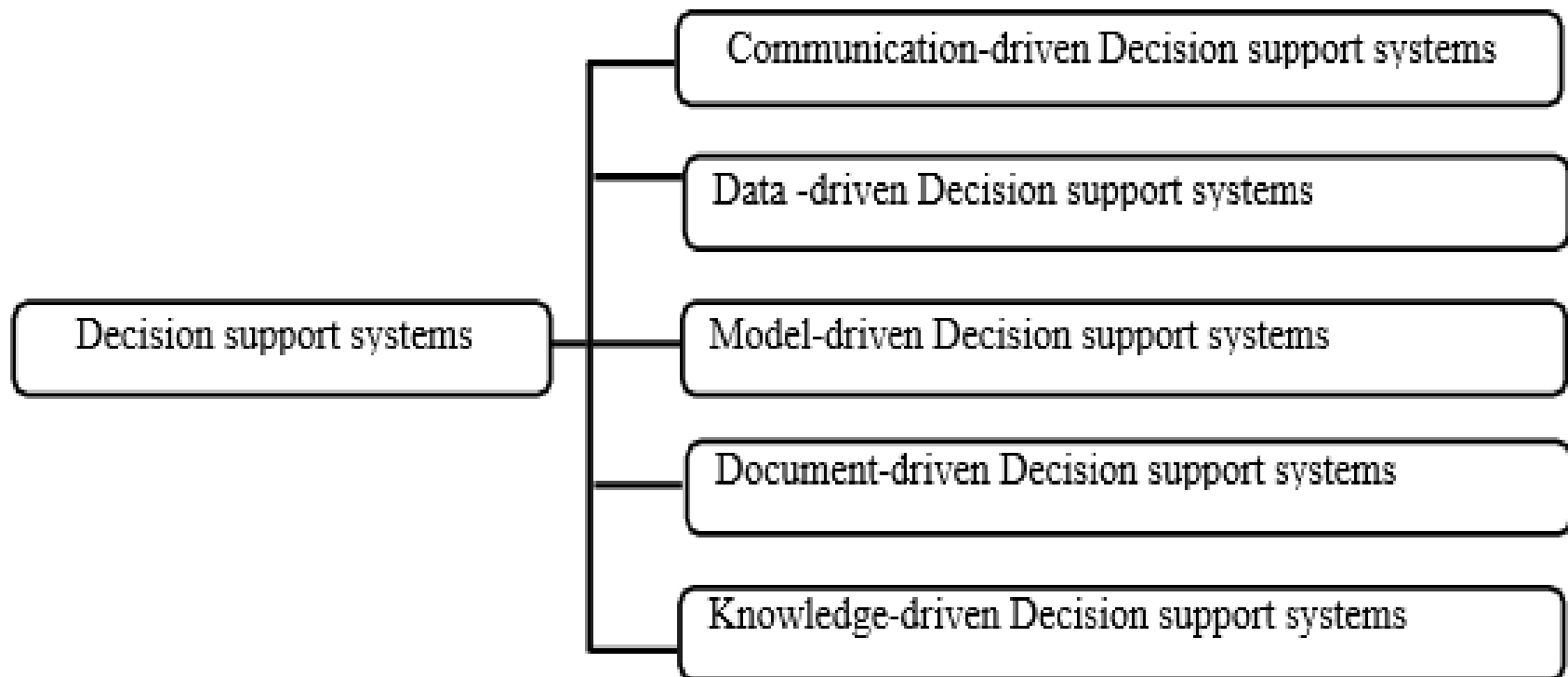
CHAPTER 3 CONTENTS

- Framework of Decisions Support Systems (DSS Classifications)
- DSS with Other Systems
- Decision-Making Levels
- Types of decisions
- Systems for Decision Support
- Four steps in decision making
- Certainty, Risk, and Uncertainty
- Decision Annalysis

REMEMBER



FRAMEWORK OF DECISIONS SUPPORT SYSTEMS (DSS CLASSIFICATIONS)



COMMUNICATION-DRIVEN DSS

- Most communications driven DSSs are targeted at **internal teams**, including partners
- Help **conduct a meeting**, or for users to **collaborate**.
- Uses network and communications technologies to facilitate collaboration and communication, and thus makes decision making faster and more productive.
- The most common technology used to deploy the DSS is a web or client server.
- Examples: chats and instant messaging software, online collaboration, and net-meeting systems.

DATA-DRIVEN DSS

- Most data driven DSSs are targeted at managers, staff and product/service suppliers.
- Supports decision making by analyzing given time-series of data and returning new information gained by those analysis
- It is used to query a database or data warehouse to seek specific answers for specific purposes.
- It is deployed via a main frame system, client/server link, or via the web.
- Examples: computer-based databases that have a query system to check (including the incorporation of data to add value to existing databases.

DOCUMENT-DRIVEN DSS

- Document-driven DSSs are more common, targeted at a broad base of user groups.
- The purpose of such a DSS is to search web pages and find documents on a specific set of keywords or search terms.
- The usual technology used to set up such DSSs are via the web or a client/server system.

KNOWLEDGE-DRIVEN DSS

- It is essentially used to provide management **advice or to choose products/services.**
- The typical deployment technology used to set up such systems could be client/server systems, the web, or software running on stand-alone PCs.

MODEL-DRIVEN DSS

- Model-driven DSSs are complex systems that help analyze decisions or choose between different options.
- These are used by managers and staff members of a business, or people who interact with the organization, for several purposes depending on how the model is set up - scheduling, decision analyses etc.
- These DSSs can be deployed via software/hardware in stand-alone PCs, client/server systems, or the web.

DECISION SUPPORT SYSTEM :TYPES (I)

1. **Traditional DSS**

Components: data management + model management
+ dialogue management + user interface

Output: decisions based on quantitative data

2. **Knowledge-based DSS (KBDSS)**

Components: DSS + Knowledge management

Output: Decisions based on quantitative &
qualitative data

DECISION SUPPORT SYSTEM :TYPES (2)

3. **Intelligent DSS based on online analytical processing (OLAP)**

Components: KBDSS + OLAP

Output: provide new hidden or
unexpected trends & patterns

4. **Group decision support systems (GDSS)**

Components: DSS + group thinking +
varying time/place facilities

Output: improve the DM process. Such
process is interactive executed by
a group of decision makers

DECISION SUPPORT SYSTEM :TYPES (3)

5. **Organizational DSS (ODSS)**

Output: improving communication and coordination among people with organizations to improve organization performance and solve problems

6. **Web-based decision support system**

Output: extend the DSS capabilities on a global basis

DSS WITH OTHER SYSTEMS (TASK REPORT)

- Transaction Processing Systems (TPS)
- Management Reporting Systems (MRS)
- Executive Information Systems (EIS)
- Office Information Systems (OIS)
- Professional Support Systems
- Expert Support systems (ESS)

BUSINESS INTELLIGENCE AND DECISION SUPPORT SYSTEMS (I)

- Because of the importance of high-quality decision making, firms are investing heavily in business intelligence systems, which consist of technologies and applications designed to help users make better business decisions.
- People's ability to combine learned knowledge with new information and change their behavior in such a way that they succeed at their task or adapt to a new situation.

BUSINESS INTELLIGENCE AND DECISION SUPPORT SYSTEMS (2)

Business intelligence (BI) is a technology-driven process for analyzing data and delivering actionable information that helps executives, managers and workers make informed business decisions.

- As part of the BI process, organizations collect data from internal IT systems and external sources, prepare it for analysis, run queries against the data and create data visualizations, BI dashboards and reports to make the analytics results available to business users for operational decision-making and strategic planning.

BUSINESS INTELLIGENCE AND DECISION SUPPORT SYSTEMS (2)



Systems and Technologies for Business Intelligence

DECISION-MAKING LEVELS (I)



DECISION-MAKING LEVELS (2)

Management support system	Needs to support	Length of time for decision-making	The nature of data & Information	Environment	type	decision / Management
IDSS / ES	+ ↑	+ ↑	- ↑	Unstructured	Strategic	Top
DSS / EIS / OR MIS				Semi Structured	Tactical	Middle
TPS OR MIS				Structured	Operational	Bottom

DECISION TYPES (I)

Unstructured

Each of these decisions is novel, important, and nonroutine, and there is no well-understood or agreed-on procedure for making them.

Structured

Repetitive and routine, and decision makers can follow a definite procedure for handling them to be efficient.

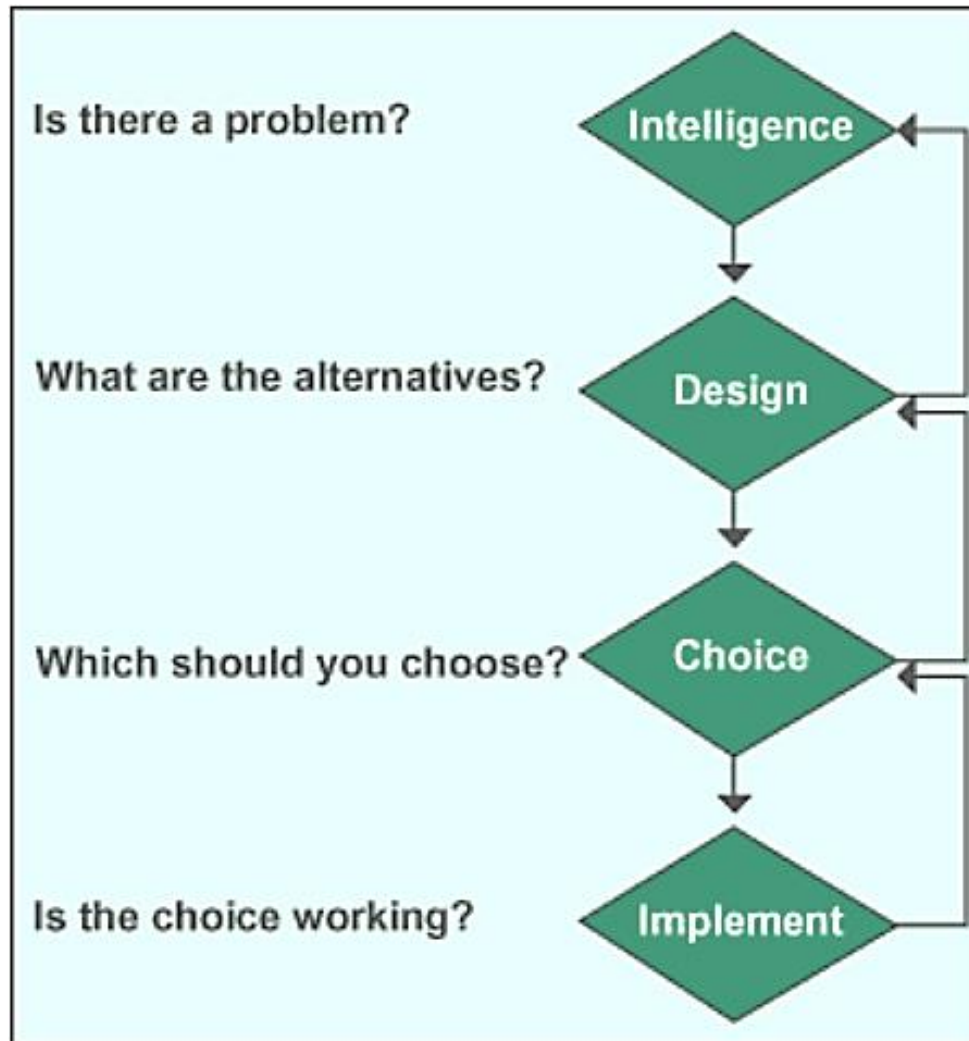
Sem-Strucured

Only part of the problem has a clear-cut answer provided by an accepted procedure

DECISION TYPES (2)

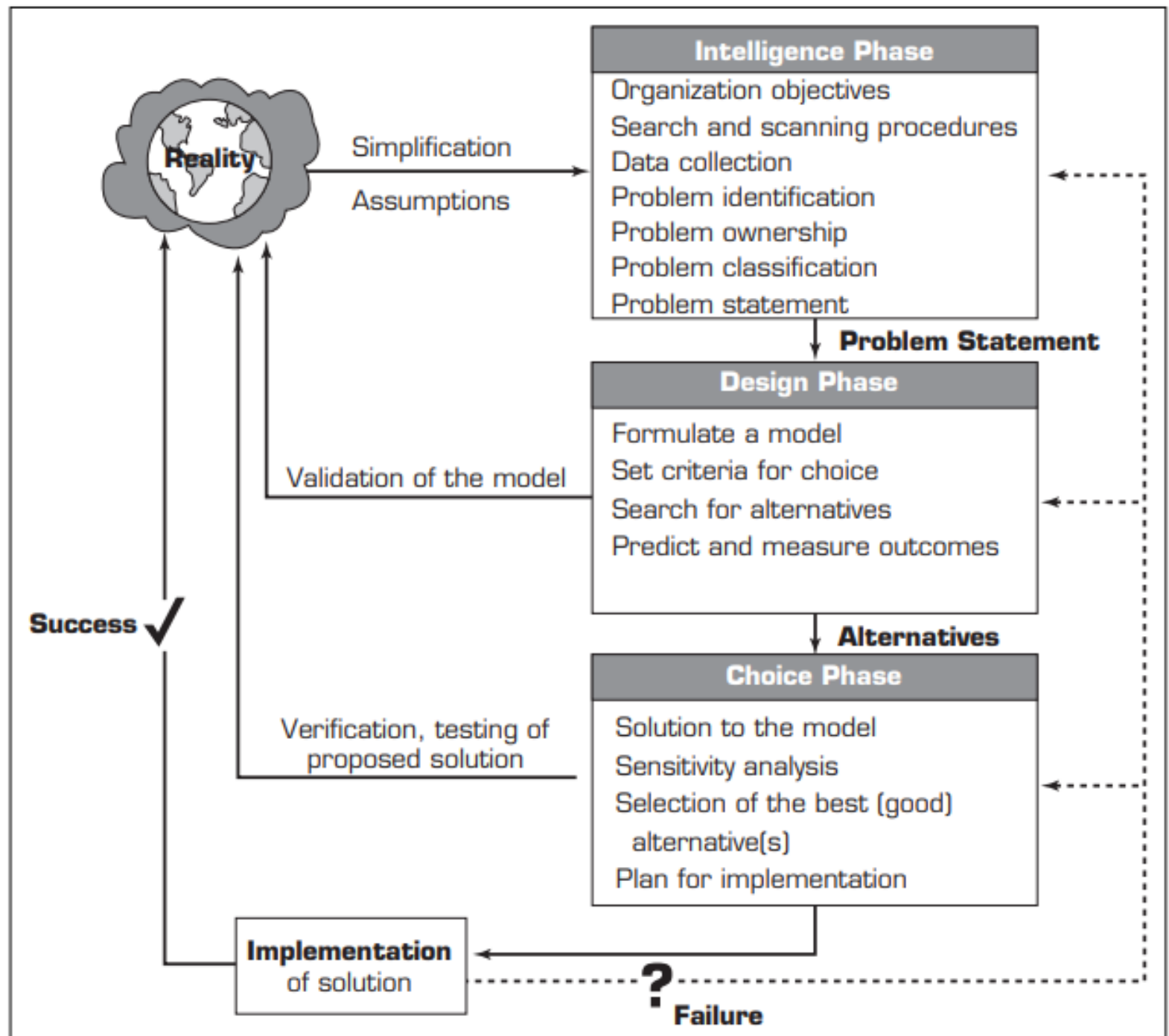
Type of Decision	Type of Control		
	Operational Control	Managerial Control	Strategic Planning
Structured	1 Accounts receivable Accounts payable Order entry	2 Budget analysis Short-term forecasting Personnel reports Make-or-buy	3 Financial management Investment portfolio Warehouse location Distribution systems
Semistructured	4 Production scheduling Inventory control	5 Credit evaluation Budget preparation Plant layout Project scheduling Reward system design Inventory categorization	6 Building a new plant Mergers & acquisitions New product planning Compensation planning Quality assurance HR policies Inventory planning
Unstructured	7 Buying software Approving loans Operating a help desk Selecting a cover for a magazine	8 Negotiating Recruiting an executive Buying hardware Lobbying	9 R & D planning New tech development Social responsibility planning

FOUR STEPS IN DECISION MAKING (I)

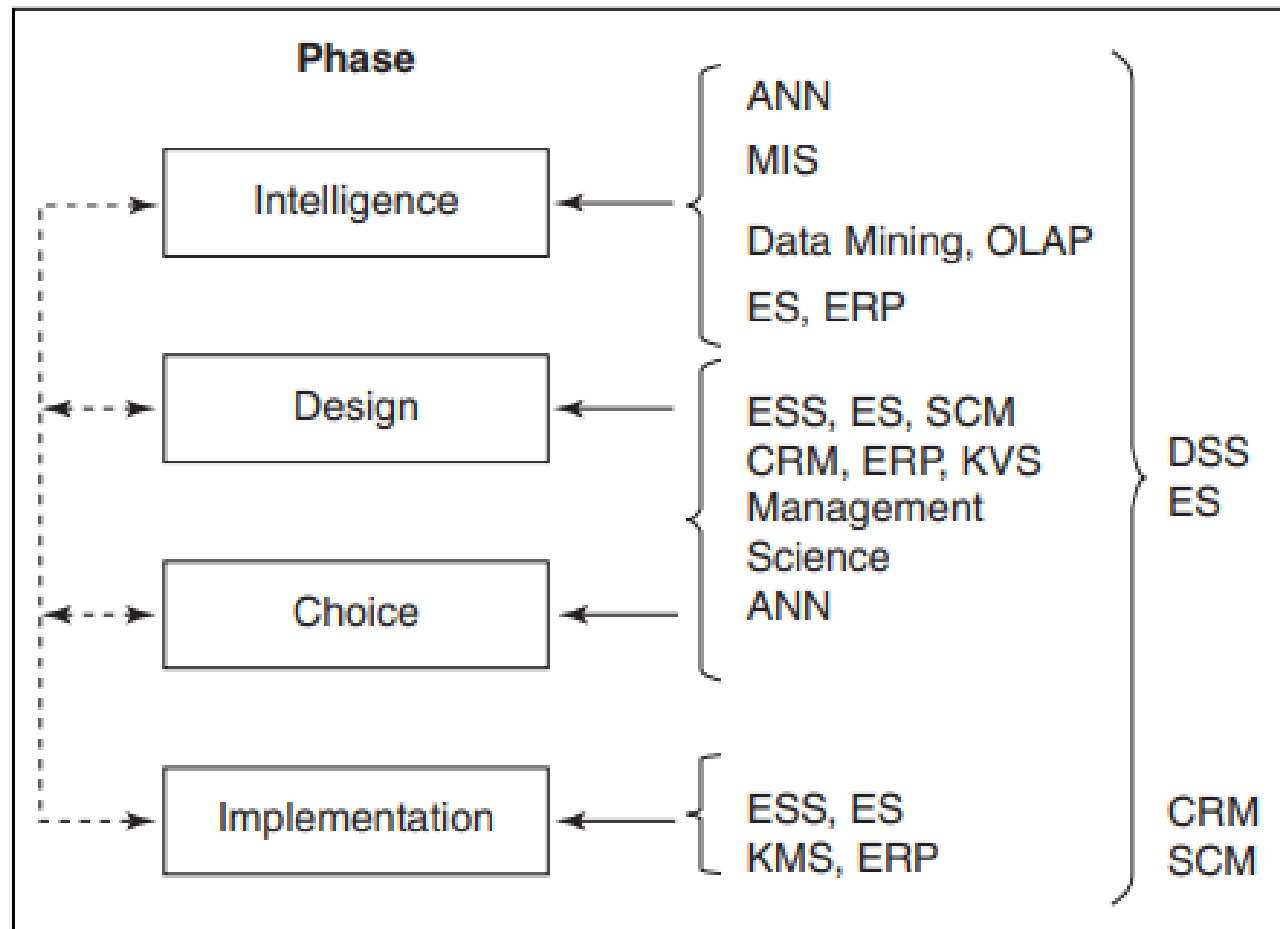


FOUR STEPS IN DECISION MAKING (2)

- **Intelligence** consists of discovering, identifying, and understanding the problems occurring in the organization—why is there a problem, where, and what effects is it having on the firm
- **Design** involves identifying and exploring various solutions to the problem.
- **Choice** consists of choosing among solution alternatives
- **Implementation** involves making the chosen alternative work and continuing to **monitor** how well the solution is working



DSS SUPPORT



DSS Support.

CERTAINTY, RISK, AND UNCERTAINTY

- Decision situations are often classified based on **what the decision-maker knows** (or believes) about the forecasted results. We classify this knowledge into three categories ranging from complete knowledge to total ignorance. These categories are:
 - Certainty
 - Risk
 - Uncertainty

DECISION-MAKING UNDER CERTAINTY

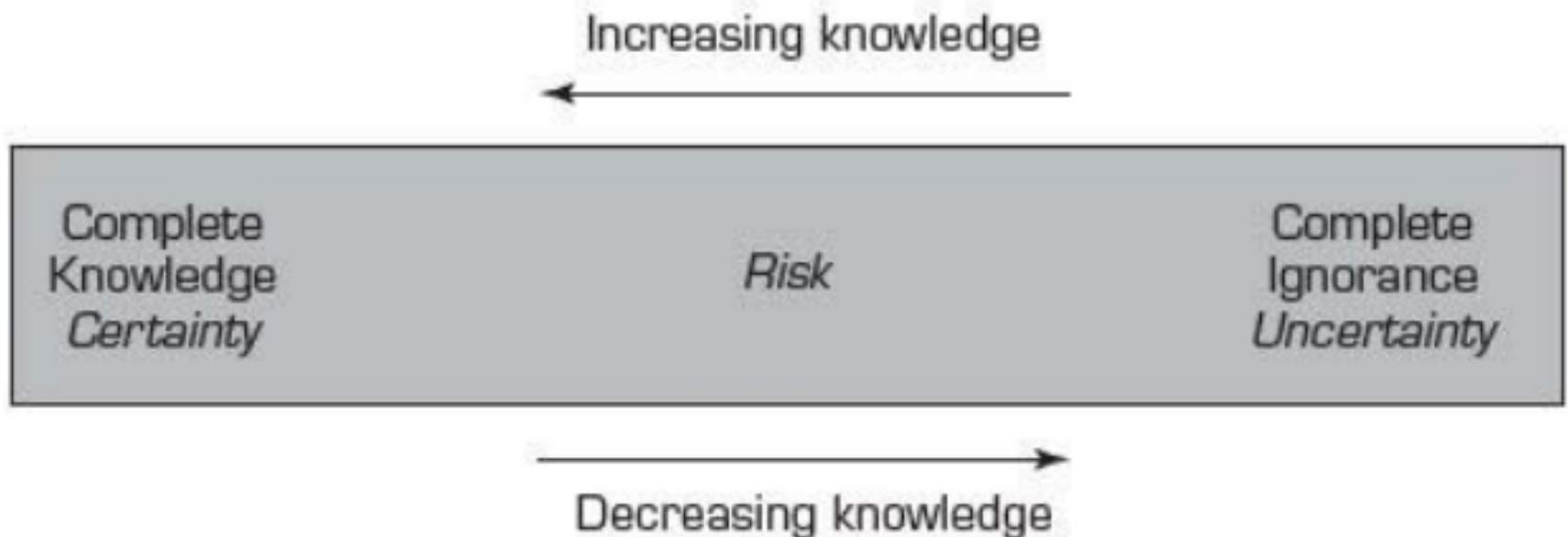
- It is assumed that **complete knowledge** is available so that the decision-maker knows exactly what the outcome of each course of action will be (as in a **deterministic** environment).
- The decision-maker is viewed as a **perfect predictor** of the future because it is assumed that there is only one outcome for each alternative
- Certainty models are relatively easy to develop and solve and can yield optimal solutions. Many financial models are constructed under assumed certainty

DECISION-MAKING UNDER UNCERTAINTY

- The decision-maker considers situations in which several outcomes are possible for each course of action.
- In this case the decision-maker does not know, or cannot estimate, the probability of occurrence of the possible outcomes.
- Decision-making under uncertainty is more difficult because of insufficient information.

DECISION-MAKING UNDER RISK

- The decision-maker must consider several possible outcomes for each alternative, each with a **given probability** of occurrence.



DECISION ANALYSIS

- **Problem Formulation**

- A decision problem is characterized by **decision alternatives**, **states of nature**, and resulting **payoffs**.
- The **decision alternatives** are the different possible strategies the decision maker can employ.
- The **states of nature** refer to future events, not under the control of the decision maker, which will ultimately affect decision results.

DECISION ANALYSIS MODELS

- **Decision theory problems are represented as one of the following:**
 - Payoff Table
 - Decision Tree

PAYOFF TABLES (I)

- The consequence resulting from a specific combination of a decision alternative and a state of nature is a **payoff**.
- **In a Payoff Table:**
 - ❖ The **rows** correspond to the possible decision **alternatives**.
 - ❖ The **columns** correspond to the possible future events (**States of Nature**).
 - ❖ The **body** of the table contains the **payoffs**.

PAYOFF TABLES (2)

THIS IS A PROFIT PAYOFF TABLE

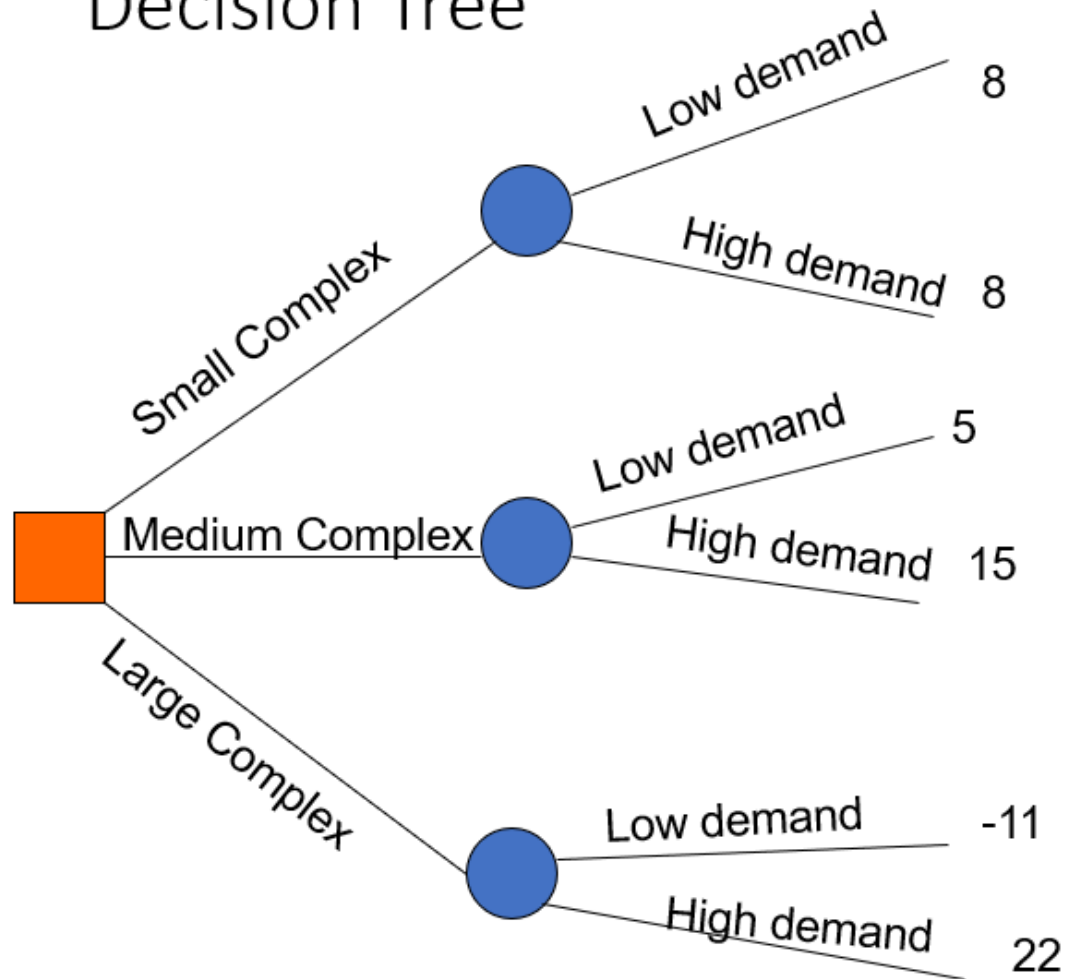
<u>Alternatives</u>	<u>States of Nature</u>	
	Low	High
Small	8	8
Medium	5	15
Large	-11	22

DECISION TREES (I)

- A **decision tree** is a graphical representation of the decision problem.
- Each decision tree has two types of nodes; **round nodes** correspond to the states of nature while **square nodes** correspond to the decision alternatives.
- The branches leaving each round node represent the different states of nature while the branches leaving each square node represent the different **decision alternatives**.
- At the end of each limb of a tree are the **payoffs** attained from the series of branches making up that limb.

DECISION TREES (2)

Decision Tree



TYPES OF DECISION-MAKING ENVIRONMENTS

- **Type 1: Decision making under certainty.**
 - Decision maker *knows* with certainty the consequences of every alternative or decision choice.
- **Type 2: Decision making under uncertainty.**
 - The decision maker *does not know* the probabilities of the various outcomes.
- **Type 3: Decision making under risk.**
 - The decision maker *knows the probabilities* of the various outcomes.

DECISION MAKING UNDER UNCERTAINTY

- **Probabilities can NOT be assigned to the occurrence of states of nature in the future. There are several methods such as:**
 - Maximax (optimistic)
 - Maximin (pessimistic)
 - Minimax regret

DECISION MAKING UNDER UNCERTAINTY

- **Maximax** (optimistic approach)
 - Choose decision with the maximum of the maximum payoffs.
- **Maximin** (conservative approach)
 - Choose decision with the maximum of the minimum payoffs.
- **Minimax regret** (regret approach)
 - Choose decision with the minimum of the maximum regrets for each alternative.

OPTIMISTIC APPROACH

- The optimistic approach would be used by an optimistic decision maker.
- This approach would be suitable for an ***optimist, or 'risk-seeking' investor***, who seeks to achieve the best results if the best happens.
- The decision with the best possible payoff is chosen.
- If the payoff table was in terms of costs, the decision with the lowest cost would be chosen.
- If the payoff table was in terms of profits, the decision with the highest profit would be chosen.

OPTIMISTIC APPROACH (EX.)

ALTERNATIVE	STATE OF NATURE		MAXIMUM IN A ROW (\$)
	FAVORABLE MARKET (\$)	UNFAVORABLE MARKET (\$)	
Construct a large plant	200,000	−180,000	200,000
Construct a small plant	100,000	−20,000	100,000
Do nothing	0	0	0

Maximax

CONSERVATIVE APPROACH

- The **conservative approach** would be used by a conservative decision maker.
- This approach would be appropriate for a ***pessimist who seeks to achieve the best results if the worst happens.***
- **If the payoff was in terms of costs**, the maximum costs would be determined for each decision and then the decision corresponding to the minimum of these maximum costs is selected. (Hence, the maximum possible cost is minimized.)
- **If the payoff was in terms of profits**, the minimum profits would be determined for each decision and then the decision corresponding to the maximum of these minimum profits is selected. (Hence, the minimum possible profit is maximized.)

CONSERVATIVE APPROACH (EX.)

ALTERNATIVE	STATE OF NATURE		MINIMUM IN A ROW (\$)
	FAVORABLE MARKET (\$)	UNFAVORABLE MARKET (\$)	
Construct a large plant	200,000	-180,000	-180,000
Construct a small plant	100,000	-20,000	-20,000
Do nothing	0	0	0

Maximin →

MINIMAX REGRET APPROACH

- The minimax regret strategy is the one that **minimizes the maximum regret. It is useful for a risk-neutral decision-maker.** Essentially, this is the technique for a 'sore loser' who does not wish to make the wrong decision.
- 'Regret' in this context is defined as the **opportunity loss or cost of having made the wrong decision**
- The minimax regret approach requires the construction of a regret table or an opportunity loss table. This is done by calculating for each state of nature the difference between each payoff and the best payoff for that state of nature.
- Then, using this regret table, the maximum regret for each possible decision is listed.
- The decision chosen is the one corresponding to the minimum of the maximum regrets.

MINIMAX REGRET APPROACH (EX. I)

■ Opportunity Loss Tables

STATE OF NATURE	
FAVORABLE MARKET (\$)	UNFAVORABLE MARKET (\$)
200,000 – 200,000	0 – (–180,000)
200,000 – 100,000	0 – (–20,000)
200,000 – 0	0 – 0

ALTERNATIVE	STATE OF NATURE	
	FAVORABLE MARKET (\$)	UNFAVORABLE MARKET (\$)
Construct a large plant	0	180,000
Construct a small plant	100,000	20,000
Do nothing	200,000	0

MINIMAX REGRET APPROACH (EX. 2)

■ Opportunity

	STATE OF NATURE	
	FAVORABLE MARKET (\$)	UNFAVORABLE MARKET (\$)
ALTERNATIVE	FAVORABLE MARKET (\$)	UNFAVORABLE MARKET (\$)
Construct a large plant	0	180,000
Construct a small plant	100,000	20,000
Do nothing	200,000	0

ALTERNATIVE	STATE OF NATURE		MAXIMUM IN A ROW (\$)
	FAVORABLE MARKET (\$)	UNFAVORABLE MARKET (\$)	
Construct a large plant	0	180,000	180,000
Construct a small plant	100,000	20,000	100,000
Do nothing	200,000	0	200,000

Minimax

DECISION MAKING UNDER RISK (I)

- Decision making when there are several possible states of nature, and we know the probabilities associated with each possible state.
- **Expected Value Approach**
 - If probabilistic information regarding the states of nature is available, one may use the expected value (EV) approach.
 - Here the expected return for each decision is calculated by summing the products of the payoff under each state of nature and the probability of the respective state of nature occurring.
- The decision yielding the best expected return is chosen

DECISION MAKING UNDER RISK (2)

- The expected value of a decision alternative is the sum of weighted payoffs for the decision alternative.
- The expected value (EV) of decision alternative d_i is defined as:

$$EV(d_i) = \sum_{j=1}^N P(s_j) V_{ij}$$

where: N = the number of states of nature

$P(s_j)$ = the probability of state of nature s_j

V_{ij} = the payoff corresponding to decision alternative d_i and state of nature s_j

EXAMPLE: BURGER PRINCE

Burger Prince Restaurant is contemplating opening a new restaurant on Main Street. It has three different models, each with a different seating capacity. Burger Prince estimates that the average number of customers per hour will be 0.4, 0.2, or 0.4. The payoff table (profits) for the three models is on the next slide.

DECISION MAKING UNDER RISK (EX. I)

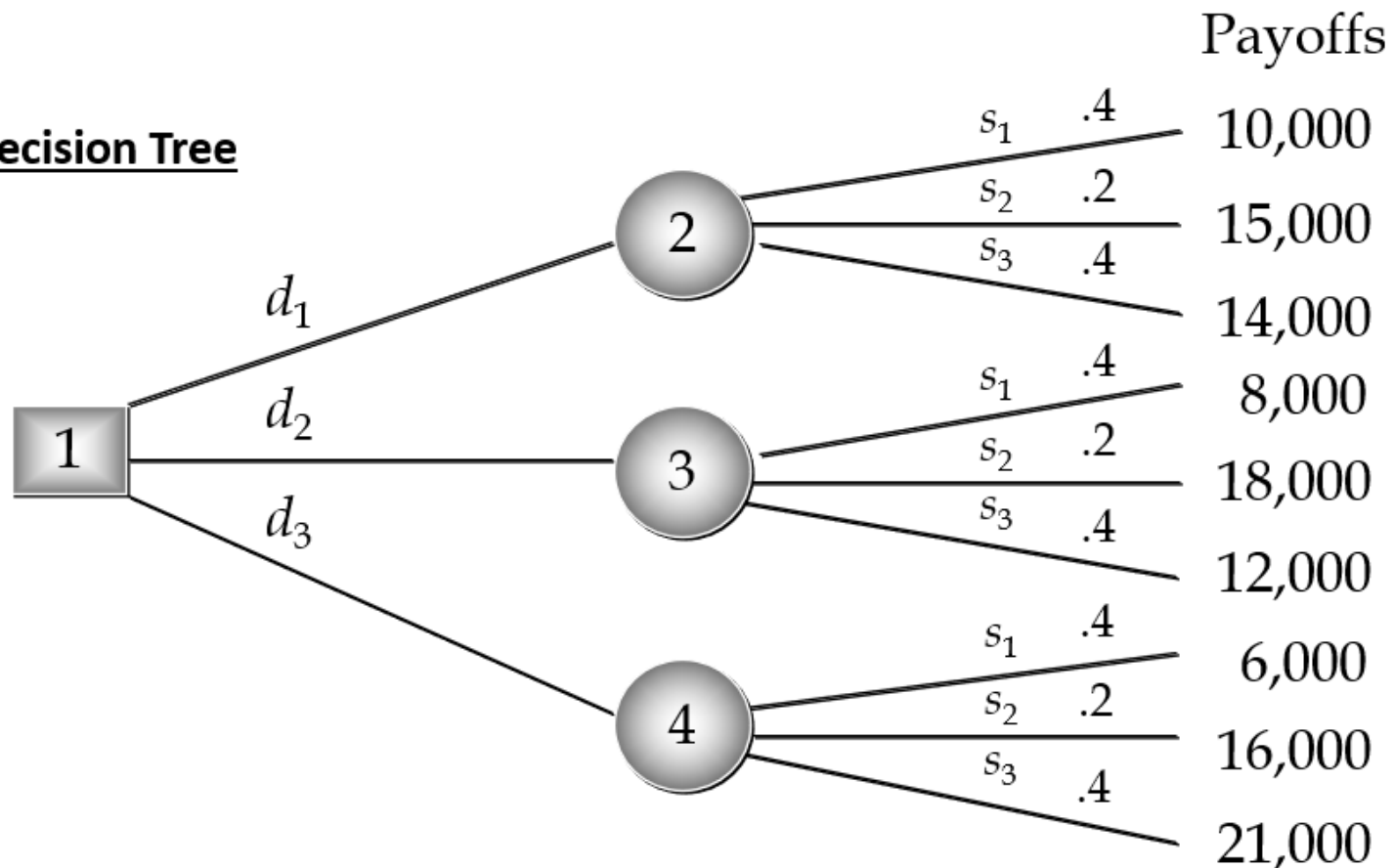
Average Number of Customers Per Hour

	$s_1 = 0.4$	$s_2 = 0.2$	$s_3 = 0.4$
Model A	\$10,000	\$15,000	\$14,000
Model B	\$ 8,000	\$18,000	\$12,000
Model C	\$ 6,000	\$16,000	\$21,000

DECISION MAKING UNDER RISK (EX. 2)

EXPECTED VALUE APPROACH

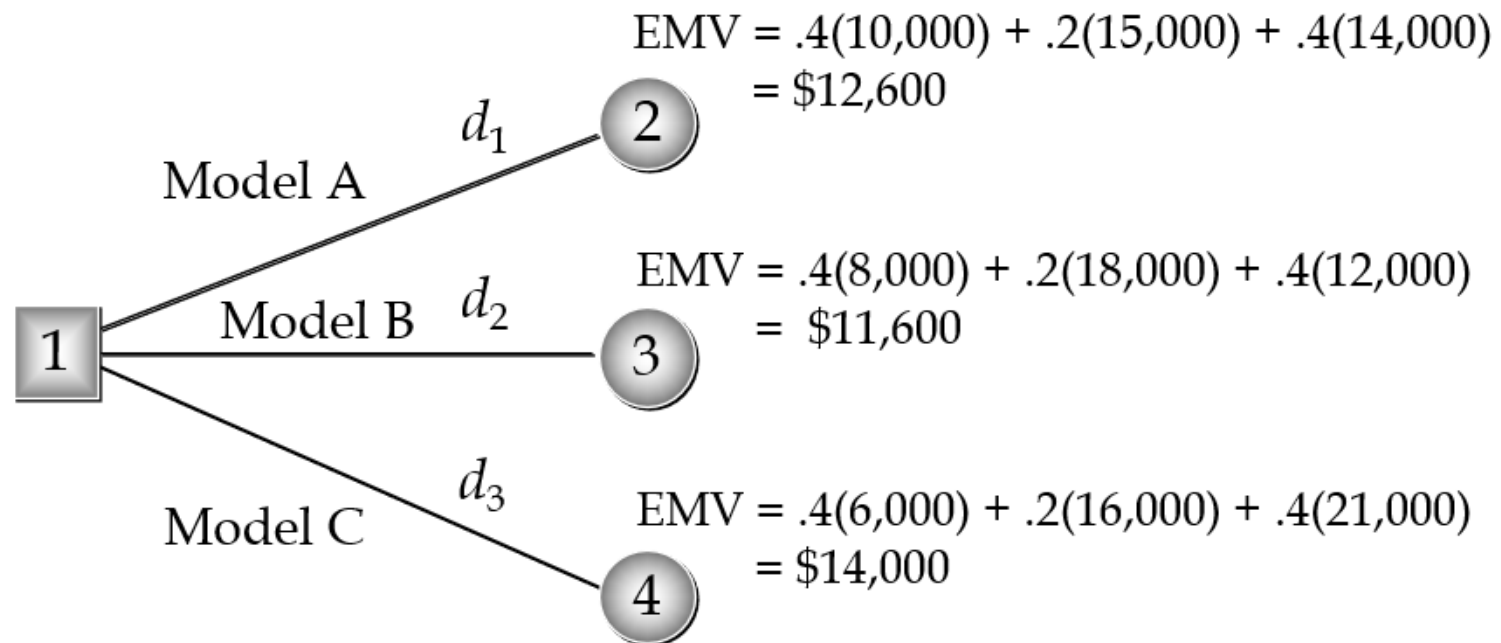
- Decision Tree



DECISION MAKING UNDER RISK (EX. 3)

EXPECTED VALUE APPROACH

⊕ Expected Value For Each Decision



Choose the model with largest EV, Model C.

DECISION MAKING UNDER RISK

EXPECTED VALUE APPROACH

■ Expected Value of Perfect Information

- Frequently information is available which can improve the probability estimates for the states of nature.
- The expected value of perfect information (EVPI) is the increase in the expected profit that would result if one knew with certainty which state of nature would occur.
- The EVPI provides an upper bound on the expected value of any sample or survey information.

DECISION MAKING UNDER RISK

EXPECTED VALUE APPROACH

- **EVPI Calculation**

- **Step 1:** Determine the optimal return corresponding to each state of nature.
- **Step 2:** Compute the expected value of these optimal returns.
- **Step 3:** Subtract the EV of the optimal decision from the amount determined in step (2).
- **Expected Value of Perfect Information**
 - Calculate the expected value for the optimum payoff for each state of nature and subtract the EV of the optimal decision.
 - $EVPI = .4(10,000) + .2(18,000) + .4(21,000) - 14,000 = \$2,000$

TASK

		States of Nature			
		Growth	Medium G	No Change	Low
		G	MG	NC	L
Actions	Bonds	12	8	7	3
	Stocks	15	9	5	-2
	Deposit	7	7	7	7

A- Determine which alternative decision maker will choose according to optimistic, pessimistic, and regret approaches

B- If the decision maker know the probabilities for each state of nature as following: 0.3,0.5,0.2, which decision will choose according to the expected value approach and compute the EVPI?



Thanks
Any questions